

ENCI335

Structural Analysis and Systems I

Tutorial 1 Solution

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Objectives

This tutorial provides an opportunity for you to practice on the following:

- Determination of degrees of statical indeterminacy (SI)
- Qualitative sketch of BMDs and deflected shapes of beams
- Analysis of statically determinate beams

Guidance on sketching deflected shape, BMD, and SFD qualitatively.

- Start with the deflected shape. Estimate the displacement and rotation at the loading points, determine the resulting rotations at nearby supports, and progressively work toward the supports further away.
- Complete the deflected shape by connecting these points to ensure a smooth curve. Remember that kinks are only permissible at internal moment hinges.
- Once the deflected shape is sketched, identify the regions of hogging and sagging moments at the supports and loading points.
- Complete the BMD by connecting these points.
- Recall that the BMD remains linear within any beam segment that does not carry a span load. It exhibits an abrupt change of slope at point load locations and a jump at point moment locations while maintaining the same slope.
- After sketching the BMD, derive the shape of the SFD, noting that the magnitude of the shear force corresponds to the slope of the bending moment diagram.
- Remember that the SFD remains constant within beam segments where no external loads are applied and exhibits a jump at point load locations.

When using the integration method starting with $\frac{dV}{dx} = w(x)$:

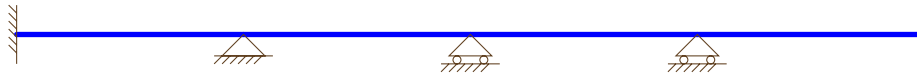
- Refer to the slides on boundary conditions to understand how to relate the forces and moments applied at the ends to the internal M and V .
- Do not include boundary forces in the load intensity function $w(x)$; these are used specifically to solve for the integration constants, C_i .
- Review the slides on Macaulay brackets for the proper procedure to represent discontinuous or point loads.

Question 1 Determination of SI

Calculate the degree of static indeterminacy (SI) of the following beams. Note, ignore the equilibrium equation and reactions in the horizontal direction. Hint: Identify first all the unknown reactions. Are the beams stable?

What would your answer for SI be if the horizontal components of equilibrium equations and reactions were included? Is there a change? Why?

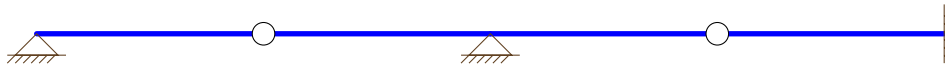
a)



Solution:

- Horizontal forces excluded:
of Reactions = 5, # of equilibrium equations = 2, **SI = 5 - 2 = 3.**
- Horizontal forces included:
of Reactions = 7, # of equilibrium equations = 3, **SI = 7 - 3 = 4.**

b)



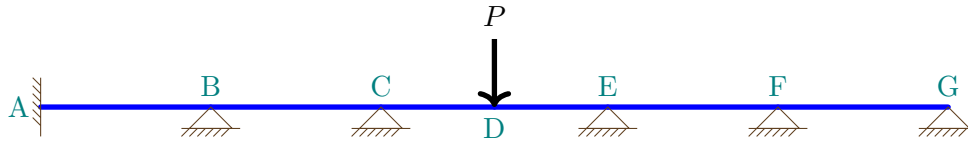
Solution:

- Horizontal forces excluded:
of Reactions = 4,
of equilibrium equations = 2,
of internal hinges = 2,
SI = 4 - 2 - 2 = 0
- Horizontal forces included:
of Reactions = 7,
of equilibrium equations = 3,
of internal hinges = 2,
SI = 7 - 3 - 2 = 2

Question 2 Deflected Shapes, BMD, and SFD

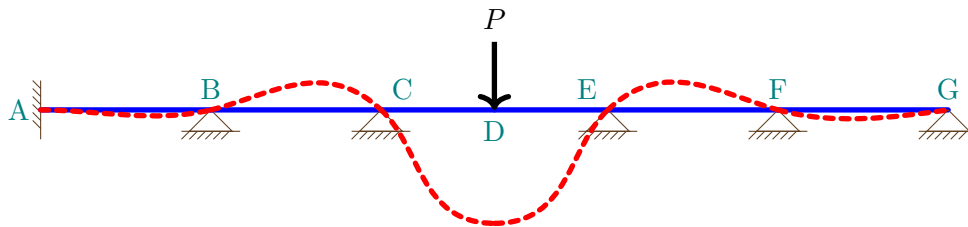
Sketch **qualitatively** the deflected shape and the bending moment diagram (BMD) and shear force diagram (SFD) of the following beams.

a)

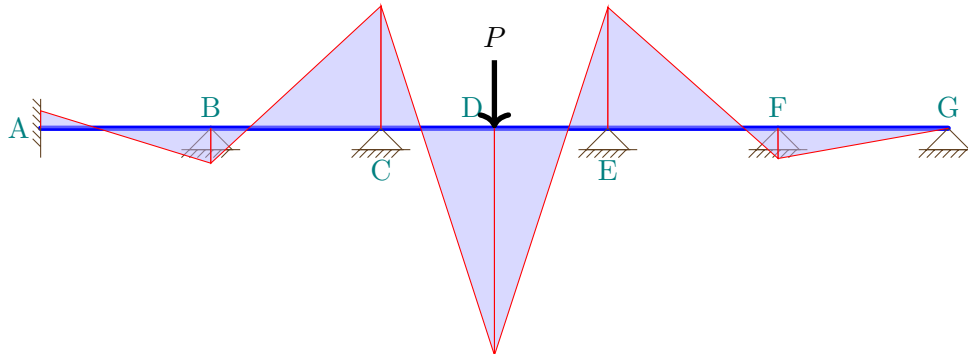


Solution:

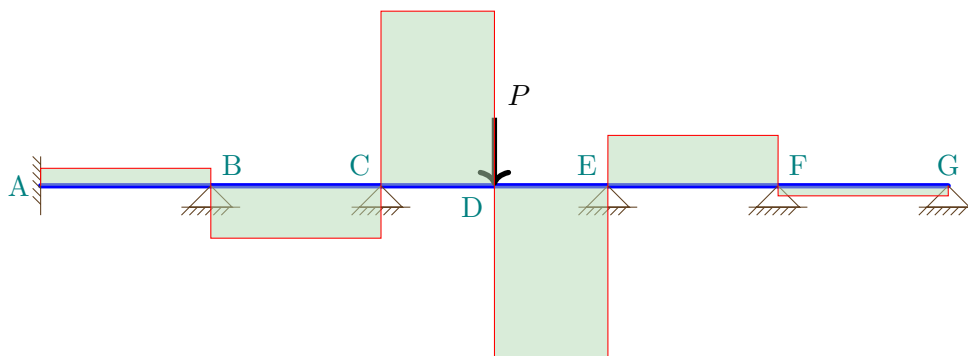
Deflected Shape:



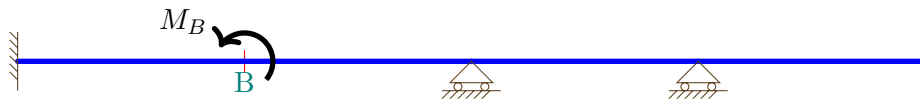
BMD



SFD

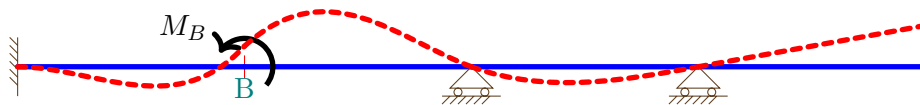


b)

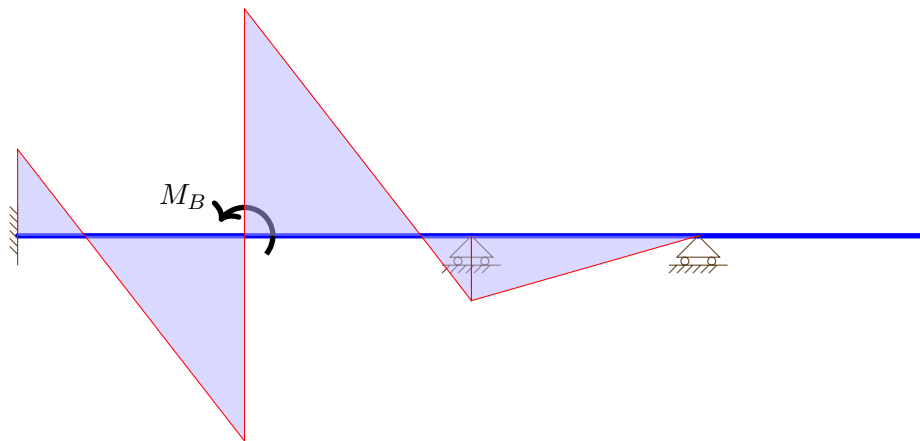


Solution:

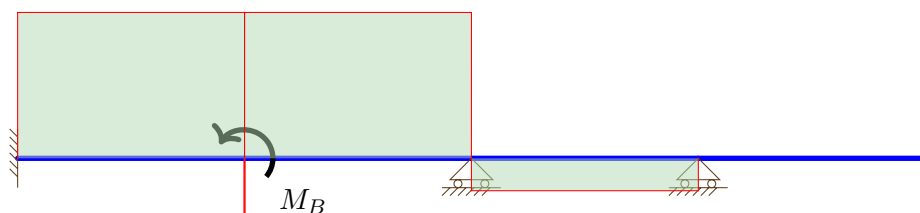
Deflected shape (the point where moment is applied moves upward a bit).



BMD (the jump at point B should be equal to the value of M_B)



SFD

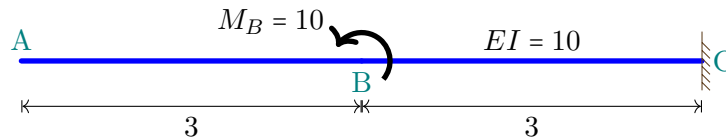


(see the explanation
in the next question)

Question 3 Analysis of beams

Calculate the support reactions of the following prismatic beams with a uniform EI , draw the BMD and SFD, and sketch qualitatively the deflected shape of the following beams. Notes, the units of dimensions and loads are not given, but they are consistent. What are the deflections and rotations at all the points labelled by A, B, C, and D, for the following beams?

a)



Solution:

This beam is statically determinate. The moment reaction at the right end, C, is -10 (clockwise). There is no vertical load applied, so no vertical reaction at C.

The bending moment $M(x)$ along the beam using Macaulay brackets is

$$M(x) = -10\langle x - 3 \rangle^0 \quad (1)$$

We can also start from the following

$$\frac{dV}{dx} = -10\langle x - 3 \rangle^{-2} \quad (2)$$

Note, the negative sign is used because the external moment applied is anti-clockwise. Integrating it once gives

$$V(x) = -10\langle x - 3 \rangle^{-1} + C_1 \quad (3)$$

Since the left end at A is a free end, there is no shear, i.e. $V(0) = 0$. It implies $C_1 = 0$.

Integrating it again gives

$$M(x) = -10\langle x - 3 \rangle^0 + C_2 \quad (4)$$

Since the left end at A is a free end, there is no moment, i.e. $M(0) = 0$. It implies $C_2 = 0$. We have recovered the same expression.

With $M(x) = EI\kappa(x)$ and $EI = 10$,

$$\kappa(x) = \frac{M(x)}{EI} = -\langle x - 3 \rangle^0 \quad (5)$$

Therefore, the beam rotation θ

$$\theta(x) = -\langle x - 3 \rangle^1 + C_3 \quad (6)$$

The rotation at C with $x = 6$ is zero, therefore,

$$\theta(6) = -3 + C_3 = 0 \implies C_3 = 3 \quad (7)$$

So, the beam rotation at A with $x = 0$ is $\theta_A = 3$.

The beam deflection v

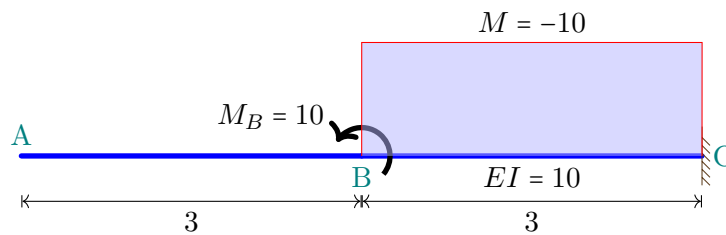
$$v(x) = -\frac{\langle x-3 \rangle^2}{2} + 3x + C_4 \quad (8)$$

The deflection at C with $x = 6$ is zero, therefore,

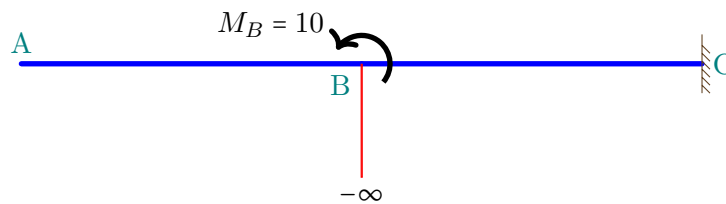
$$v(6) = -9/2 + 18 + C_4 = 0 \implies C_4 = -27/2 = -13.5 \quad (9)$$

The deflection at A with $x = 0$ is $v_A = -13.5$.

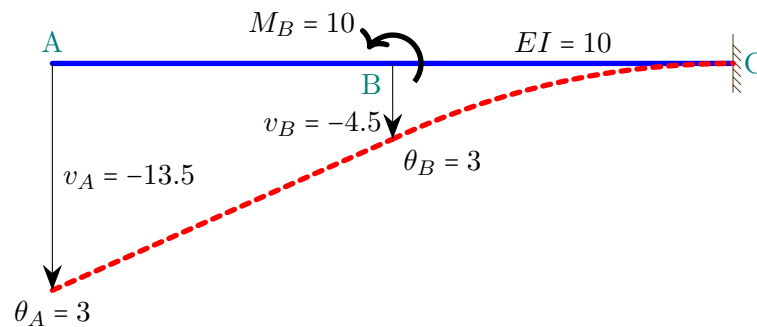
BMD



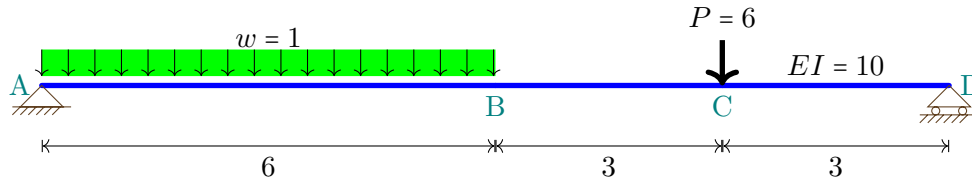
SFD: By taking the derivative of M with respect to x , we have $V(x) = -10\langle x-3 \rangle^{-1}$. This expression means, at the point ($x = 3$) where the point moment is applied, the value of shear is infinite in the negative direction, and is zero everywhere else.



Deflection:



b)



Solution:

Using the moment and force equilibrium equations, the left and right reaction forces can be obtained as 6 (upwards).

The bending moment $M(x)$ can be derived as

$$M(x) = 6x - \frac{x^2}{2} + \frac{\langle x - 6 \rangle^2}{2} - 6\langle x - 9 \rangle^1 \quad (10)$$

Taking the derivative of M with respect to x gives the shear force $V(x)$ as

$$V(x) = 6 - x + \langle x - 6 \rangle^1 - 6\langle x - 9 \rangle^0 \quad (11)$$

With $EI = 10$, the beam curvature $\kappa(x)$ is

$$10\kappa(x) = 6x - \frac{x^2}{2} + \frac{\langle x - 6 \rangle^2}{2} - 6\langle x - 9 \rangle^1 \quad (12)$$

Integrate the curvature with respect to x gives the beam rotation θ

$$10\theta(x) = 3x^2 - \frac{x^3}{6} + \frac{\langle x - 6 \rangle^3}{6} - 3\langle x - 9 \rangle^2 + C_1 \quad (13)$$

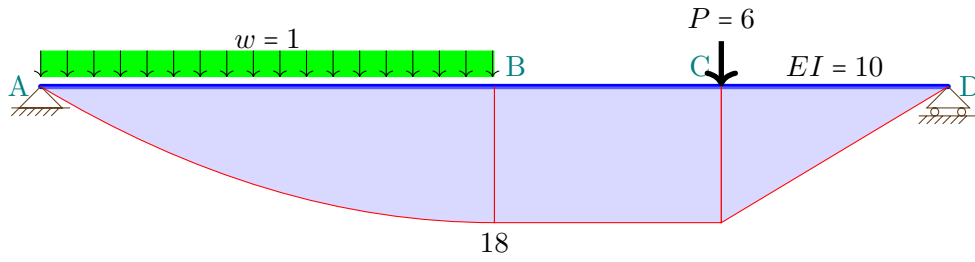
Integrate the rotation with respect to x gives the beam deflection v

$$10v(x) = x^3 - \frac{x^4}{24} + \frac{\langle x - 6 \rangle^4}{24} - \langle x - 9 \rangle^3 + C_1x + C_2 \quad (14)$$

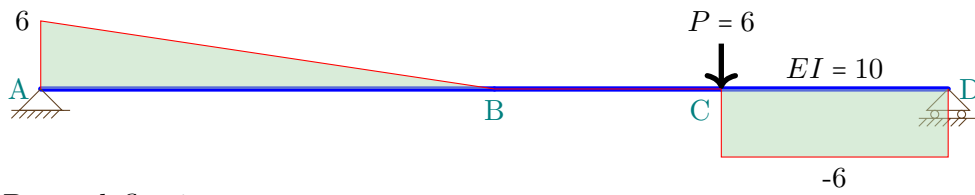
The beam deflections at both ends are zero, therefore $C_1 = -74.25$, $C_2 = 0$.

$$10v(x) = x^3 - \frac{x^4}{24} + \frac{\langle x - 6 \rangle^4}{24} - \langle x - 9 \rangle^3 - 74.25x \quad (15)$$

BMD



SFD



Beam deflection

